17 Simplify fully $\dfrac{3x^2 - 8x - 3}{2x^2 - 6x}$

Numerator: $3 \times (-3) = -9$. Split $-8x$ into $-9x + x$. Denominator: HCF = $2x$

×	x	-3
$3x$	$3x^2$	$-9x$
1	x	-3

1 mark
Factorise a numerator or denominator

×	x	-3
$2x$	$2x^2$	$-6x$

$3x^2 - 8x - 3 = (3x + 1)(x - 3)$

1 mark
Both factorisations correct

$2x^2 - 6x = 2x(x - 3)$

$$\frac{(3x + 1)(x - 3)}{2x(x - 3)} = \frac{3x + 1}{2x}$$

1 mark
Cancel the whole factor x – 3

Retain the original restrictions: $x \neq 0$ and $x \neq 3$.

$\dfrac{3x + 1}{2x}$

(Total for Question 17 is 3 marks)

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